



Team 1
Project:
Parking Lot System



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Feature 1:

Entry/Exit Management and Manual Space Allocation

Manages the registration of vehicle movements and the manual update of parking spot availability within the database.

- **Manual Entry/Exit Logging:** Allows the operator to record the plate and time when a vehicle enters or leaves the system.
- **Database Spot Update:** Updates the "occupied" or "available" status of a specific ID in the database once the operator confirms a movement.
- **License Plate Validation:** Enforces a specific text pattern (ABC-1234) for every record to ensure data consistency.
- **Occupancy Table View:** Provides a list of available parking spots for the operator to choose from when registering a new vehicle.

Feature 2:

Resident Rental and Subscription Management

Handles the administrative data for residents, their assigned vehicles, and their specific rental payment status.

- **Resident Data Administration:** Stores and manages information such as name, apartment number, and contact details for each resident.
- **Rental Status Tracking:** Monitors whether a resident's account is in "Pending," "Paid," or "Canceled" status.
- **Subscription Renewal:** Allows for the extension of rental periods (from 1 to 12 months) manually selected by the user or admin.
- **Vehicle-Resident Association:** Links specific vehicle plates to a resident's profile to manage authorized parking rights.



Feature 3:

Data Reporting and PDF Generation

Converts system data into readable tables and official documents for administrative review.

- **General Vehicle List:** Displays a complete table of all vehicles registered in the system and their current parking status.
- **Automated PDF Export:** Features a tool to capture the data from the screen and generate a professional PDF document.
- **Historical Records Table:** Shows a log of previous entries and exits stored in the "Entrances" collection for auditing purposes.
- **Visitor Logs:** Lists all registered visitors along with the resident who authorized their presence.

Feature 4:

System Access and Visitor Registration

Controls software access via user roles and manages temporary permissions for external guests.

- **Role-Based Login:** Provides separate access for "Security Guards" and "Residents" using basic credential verification.
- **Visitor Authorization:** Allows for the registration of guests, linking them to a host resident ID to keep a record of who is visiting whom.
- **Entry Pass Tracking:** Manages a simple boolean check to indicate if a visitor has been granted an entry pass.
- **Real-Time Database Connectivity:** Ensures the application is actively connected to the cloud database (MongoDB) before performing any operation.